

19 Ans: A

At same operating voltage,

$$P_Q = 4P_P$$
$$V^2 / R_Q = (4)(V^2 / R_P)$$
$$R_P = 4R_Q$$

When placed in series, current through P and Q is the same.

Hence,

$$P_Q' = I^2 R_Q$$
$$P_P' = I^2 R_P = I^2 (4R_Q)$$

$$P_P' = 4P_Q'$$

20 Ans: B